

Guransh Singh Arora

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Curriculum Vitae

Profile Summary

Passionate Mechatronics Engineering student specializing in robotics, embedded systems, and intelligent hardware design. Experienced in developing autonomous robots, UAVs and secure IoT systems by combining cross-disciplinary engineering with a hands-on, problem-solving mindset.

Education

B.Tech., Mechatronics | Manipal University Jaipur, Rajasthan | Last Sem GPA: 9.42 | CGPA: 8.44

Class XII | CBSE, Indraprastha World School, Delhi | Percentage: 85.4% 2022-23

Class X | CBSE, Indraprastha World School, Delhi | Percentage: 84.1% 2020-21

Experience

1. **Head, Projects and Research, ACM SIGBED Chapter** – Mentored and supervised student research in embedded and cyber-physical systems (2024-25).
2. Delivered student-led lectures on **MATLAB Simulink** and **PID control systems** (2025).
3. Selected for **Erasmus+ research exchange program** (2025-26).

Patents & Research

1. **Smart Parcel Box** project as **Intelligent Parcel Delivery Box with Authenticated Access and Theft Detection System** — Patent published in India (Application No. [202511076582](#)) and granted in Germany (Patent No. [202025104999](#))
2. **Indo-German Challenge for Sustainable Production of Photovoltaic Modules**, funded by **Erasmus+ Program**
 - Selected to represent **Manipal University Jaipur** in a **collaborative research program** with **Technische Universität Braunschweig (Germany)** under the Erasmus+ initiative.
 - Engaged in research on **sustainable fabrication methods and performance optimization** of photovoltaic (PV) modules for enhanced energy efficiency and durability.
 - Collaborating with international teams to develop **eco-friendly manufacturing strategies** aligning with global renewable energy and sustainability goals.

Projects

1. **Fixed-Wing UAV Development, Project-Based Learning I, Manipal University Jaipur, Rajasthan**
 - Built a **Trainer RC plane**, applying aerodynamics and electronics integration for stable flight.
 - Integrated **1000kv brushless motor**, **SG90 servos**, **3s LiPo battery**, and a **Flysky transmitter-receiver system** for remote-controlled operation.
 - Conducted **ground testing**, ensuring **structural integrity**, **weight balance**, and **control responsiveness** for future flight trials.

2. Digital Timer using 8051 Microcontroller, Manipal University Jaipur, Rajasthan

- Designed and implemented a **countdown timer** using the AT89S52 microcontroller with a **4-digit 7-segment display** and **multiplexing techniques** for efficient output control.
- Integrated **TTP223 touch sensor** for modern touch-based start/stop functionality and **push buttons** for precise minute/second adjustments.
- Developed software in **C** using **Keil uVision IDE**, incorporating **debouncing mechanisms** for reliable user input and a **buzzer alarm system** for countdown completion.
- Ensured **high timing accuracy (<1% deviation)** with a **12 MHz crystal oscillator**, optimized **power efficiency**, and conducted **extensive testing** for stability and performance.

3. Smart Parcel Box, Project-Based Learning II, Manipal University Jaipur, Rajasthan

- Designed a fully functional smart parcel box in **Fusion 360** and fabricated using **PLA+** on a **Creality 3D printer**.
- Powered by an **ESP32**, it includes a **barcode scanner**, **keypad**, **IR sensors**, **vibration sensors**, and a **locking mechanism** for secure parcel delivery.
- Features **barcode scanning for parcel authentication**, **auto-locking**, **OTP display**, **theft detection**, and **Wi-Fi-enabled notifications** using **Blynk IoT** app, integrating real-time tracking and smart access control.

4. Go-to-Goal Mobile Robot, Manipal University Jaipur, Rajasthan (Mentor Funded Project)

- Designed and built an autonomous robot using **Arduino**, **MPU6050**, and **encoder-based TT motors** for dead-reckoning motion.
- Used **Bluetooth (HC-05)** to input target coordinates and optional way point coordinates, enabling the robot to navigate in the X-Y plane from an initial origin point (0,0).
- Implemented Polar Coordinate control to drive the robot from one coordinate to another in a smooth trajectory and cascaded along a PID controller to implement the desired wheel velocities given by the Polar Coordinate controller.

5. Gear Test Rig, Manipal University Jaipur, Rajasthan (Ongoing – University Funded Project)

- Developing a novel test rig for evaluating the **torque capacity**, **efficiency**, and **failure limits** of mechanical gears.
- Designed, assembled, and simulated all mechanical components using **Autodesk Fusion 360** with dynamic motion analysis.
- Currently overseeing **fabrication and electronics integration**, including sensor instrumentation and data acquisition for automated testing and analysis.

6. Ornithopter: Bio-Inspired flapping wing drone, Manipal University Jaipur, Rajasthan (University Funded Project)

- Developing a **bio-inspired aerial robot** that mimics avian flight dynamics through flapping-wing actuation for efficient lift and maneuverability.
- Currently focusing on **aerodynamic modeling**, **material selection**, and **mechanism design** using **Autodesk Fusion 360** for structural and motion analysis.

- Aiming to achieve **stable flight and hover capability** using lightweight composite structures and an **optimized flapping mechanism**.

Honours & Awards

- **Dean's List Award:** For scoring 9.24/10 GPA in IVth semester
- **2nd Position in Department's Project Expo:** For Smart Parcel Box

Technical Skills

- **Embedded Systems:** 8051, Arduino, ESP32, Sensors & Actuators, Keil μ Vision
- **CAD Modelling:** Fusion 360, AutoCAD
- **Programming:** C, Embedded C, Python
- **Simulation & Modeling:** MATLAB, Simulink
- **Electronics:** EasyEDA, Circuit Design, Debugging

Core Strengths

- **Problem-Solving & Critical Thinking**
- **Project Management and Team Leadership**